



Session 07: Solution Stoichiometry & Molarity

Topic: When reactions happen in water, concentration is the new mass.

Time: 150 minutes

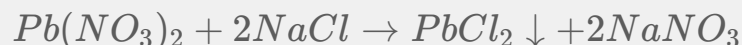
Big Fat Notebook: Chapters 8, 10

Opening Attack

Try this:

You add **35.0 mL of 0.250 M $Pb(NO_3)_2$ (lead(II) nitrate)** to **50.0 mL of 0.180 M $NaCl$ (sodium chloride)**.

A white precipitate of $PbCl_2$ (lead(II) chloride) forms:



- (a) How many moles of Pb^{2+} are in the first solution?
- (b) How many moles of Cl^- are in the second solution?
- (c) Which is the limiting reagent?
- (d) How many **grams** of $PbCl_2$ precipitate?

Stuck? Good. By the end of this session, you'll convert solution volumes and concentrations to moles, handle dilution, and solve precipitation stoichiometry — all in your head.

Worked Pattern

Example 1 — Molarity from Mass and Volume

Problem: You dissolve 5.84 g of NaCl (sodium chloride) in enough water to make 250.0 mL of solution. What is the molarity?

Step 1: Convert mass to moles.

$$\text{Molar mass NaCl} = 22.99 + 35.45 = 58.44 \text{ g/mol}$$

$$\text{mol NaCl} = 5.84 \text{ g} \div 58.44 \text{ g/mol} = 0.09993 \text{ mol}$$

Step 2: Convert volume to liters.

$$250.0 \text{ mL} = 0.2500 \text{ L} \text{ (This is the \#1 place students lose points — forgetting this conversion!)}$$

Step 3: Divide moles by liters.

$$M = \frac{\text{mol}}{\text{L}} = \frac{0.09993 \text{ mol}}{0.2500 \text{ L}} = 0.3997 \text{ M}$$

Answer: 0.400 M NaCl (3 sig figs, limited by 5.84).

$$M = \frac{\text{mol}}{\text{L}} \quad \text{mol} = M \times V(\text{L}) \quad V(\text{L}) = \frac{\text{mol}}{M}$$

Example 2 — Moles from Molarity and Volume

Problem: How many moles of H_2SO_4 (sulfuric acid) are in 35.0 mL of 0.250 M H_2SO_4 ?

Step 1: Convert mL to L.

$$35.0 \text{ mL} = 0.0350 \text{ L}$$

Step 2: Multiply molarity by liters.

$$\begin{aligned} \text{mol } H_2SO_4 &= M \times V = 0.250 \text{ mol/L} \times 0.0350 \text{ L} \\ &= 0.00875 \text{ mol} \end{aligned}$$

Answer: $8.75 \times 10^{-3} \text{ mol } H_2SO_4$.

Example 3 — Dilution: $M_1 V_1 = M_2 V_2$

Problem: You have 50.0 mL of 2.00 M HCl. You add water until the total volume is 200.0 mL. What is the new concentration?

Step 1: Identify knowns.

$$M_1 = 2.00 \text{ M}, V_1 = 50.0 \text{ mL}, V_2 = 200.0 \text{ mL}$$

Step 2: Apply $M_1 V_1 = M_2 V_2$.

$$M_2 = \frac{M_1 V_1}{V_2} = \frac{2.00 \text{ M} \times 50.0 \text{ mL}}{200.0 \text{ mL}}$$

Step 3: Solve.

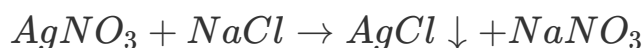
$$M_2 = \frac{100.0}{200.0} = 0.500 \text{ M}$$

Answer: 0.500 M HCl .

Key insight: $M_1V_1 = M_2V_2$ works because **moles don't change during dilution** — you're just spreading the same number of moles through more water. The units of V can be mL (they cancel) as long as both are the same unit.

Example 4 — Solution Stoichiometry (Simple)

Problem: How many grams of $AgCl$ (silver chloride) precipitate when 25.0 mL of 0.100 M $AgNO_3$ (silver nitrate) reacts with excess NaCl?



Step 1: mol $AgNO_3$ from M and V .

$$\text{mol } AgNO_3 = 0.100 \text{ M} \times 0.0250 \text{ L} = 0.00250 \text{ mol}$$

Step 2: Mole ratio to product.

From equation: $1 \text{ mol } AgNO_3 \rightarrow 1 \text{ mol } AgCl$

$$\text{mol } AgCl = 0.00250 \text{ mol}$$

Step 3: Moles to grams.

$$\text{Molar mass } AgCl = 107.87 + 35.45 = 143.32 \text{ g/mol}$$

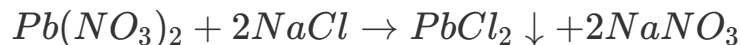
$$\text{g } AgCl = 0.00250 \text{ mol} \times 143.32 \text{ g/mol} = 0.3583 \text{ g}$$

Answer: $0.358 \text{ g } AgCl$.

$$\text{mL} \xrightarrow{\div 1000} \text{L} \xrightarrow{\times M} \text{mol} \xrightarrow{\times \text{mole ratio}} \text{mol product} \xrightarrow{\times \text{MM}} \text{g product}$$

Example 5 — Precipitation with Limiting Reagent

Problem: 35.0 mL of 0.250 M $Pb(NO_3)_2$ + 50.0 mL of 0.180 M NaCl. How many grams of $PbCl_2$ precipitate?



Step 1: Moles of each reactant.

$$\text{mol } Pb(NO_3)_2 = 0.250 \times 0.0350 = 0.00875 \text{ mol}$$

$$\text{mol NaCl} = 0.180 \times 0.0500 = 0.00900 \text{ mol}$$

Step 2: Find limiting reagent.

Divide by coefficients:

$$Pb(NO_3)_2 \text{ ratio} = 0.00875/1 = 0.00875$$

$$\text{NaCl ratio} = 0.00900/2 = 0.00450$$

$$0.00450 < 0.00875 \rightarrow \text{NaCl is limiting.}$$

Step 3: Theoretical yield from limiting reagent.

$$\text{mol } PbCl_2 = 0.00900 \text{ mol NaCl} \times \frac{1 \text{ mol } PbCl_2}{2 \text{ mol NaCl}} = 0.00450 \text{ mol } PbCl_2$$

$$\text{Molar mass } PbCl_2 = 207.2 + 2(35.45) = 278.10 \text{ g/mol}$$

$$\text{g } PbCl_2 = 0.00450 \times 278.10 = 1.2515 \text{ g}$$

Answer: 1.25 g $PbCl_2$ precipitate (3 sig figs).

This is the opening problem — solved.

THE TRAP — Forgetting to Convert mL to L

The wrong way:

50.0 mL of 0.180 M NaCl. "mol = $0.180 \times 50.0 = 9.00 \text{ mol.}$ " ← **WRONG!**

Why is this wrong? Molarity is **moles per liter**, not moles per milliliter. 50.0 mL = 0.0500 L, not 50.0 L. Multiplying by mL gives an answer 1000× too large.

The right way:

mol NaCl = $0.180 \text{ mol/L} \times 0.0500 \text{ L} = 0.00900 \text{ mol.}$ ✓

Spot the difference: $0.180 \times 50.0 = 9.00$ (wrong, off by 1000×) vs. $0.180 \times 0.0500 = 0.00900$ (right).

Mental checkpoint: If your answer says there are 9 moles of something in a 50 mL test tube, that's physically absurd — 9 mol of NaCl would be about 526 g, more than half a kilogram in a tiny tube!

Example 6 — Ion Concentrations in Solution

Problem: What is the concentration of each ion in 0.200 M Na_2SO_4 (sodium sulfate)?

Step 1: Write the dissociation equation.



Step 2: Use coefficients as multipliers.

$$[\text{Na}^+] = 2 \times 0.200 \text{ M} = 0.400 \text{ M}$$

$$[\text{SO}_4^{2-}] = 1 \times 0.200 \text{ M} = 0.200 \text{ M}$$

Answer: $[Na^+] = 0.400\text{ M}$, $[SO_4^{2-}] = 0.200\text{ M}$.

Example 7 — Preparing a Solution

Problem: How many grams of $CuSO_4 \cdot 5H_2O$ (copper(II) sulfate pentahydrate) are needed to prepare 500.0 mL of 0.100 M $CuSO_4$?

Step 1: Moles of $CuSO_4$ needed.

$$\text{mol } CuSO_4 = 0.100\text{ M} \times 0.5000\text{ L} = 0.0500\text{ mol}$$

Step 2: Molar mass of the hydrate.

$CuSO_4 \cdot 5H_2O$:

$$\text{Cu} = 63.55, \text{S} = 32.07, \text{O}_4 = 64.00, 5H_2O = 5 \times 18.016 = 90.08$$

$$\text{Total} = 63.55 + 32.07 + 64.00 + 90.08 = 249.70\text{ g/mol}$$

Step 3: Mass needed.

$$\text{g} = 0.0500\text{ mol} \times 249.70\text{ g/mol} = 12.485\text{ g}$$

Answer: $12.5\text{ g } CuSO_4 \cdot 5H_2O$.

Important: When a hydrated salt is used, the water molecules are part of the molar mass! Don't forget them.

Example 8 — Mixing Two Solutions: New Concentration

Problem: You mix 150.0 mL of 0.400 M NaCl with 250.0 mL of 0.200 M NaCl. What is the

$[NaCl]$ in the mixture? What is $[Na^+]$?

Step 1: Total moles of NaCl.

From first solution: $\text{mol} = 0.400 \times 0.1500 = 0.0600 \text{ mol}$

From second solution: $\text{mol} = 0.200 \times 0.2500 = 0.0500 \text{ mol}$

Total mol NaCl = $0.0600 + 0.0500 = 0.1100 \text{ mol}$

Step 2: Total volume.

$V_{\text{total}} = 150.0 + 250.0 = 400.0 \text{ mL} = 0.4000 \text{ L}$

Step 3: New concentration.

$[NaCl] = \frac{0.1100 \text{ mol}}{0.4000 \text{ L}} = 0.2750 \text{ M}$

Since NaCl dissociates completely: $[Na^+] = [Cl^-] = 0.2750 \text{ M}$

Answer: $[NaCl] = 0.275 \text{ M}, [Na^+] = 0.275 \text{ M}, [Cl^-] = 0.275 \text{ M}.$

What We Just Did — The Solution Stoichiometry Procedure

Step 1: Convert solution information to MOLES.

If given **molarity and volume**: $\text{mol} = M \times V(\text{L})$.

Always convert mL to L first — divide by 1000.

If given **mass and volume**: $\text{mol} = \text{g} \div \text{molar mass}$, then $M = \text{mol}/V(\text{L})$.

Step 2: Use the mole ratio from the balanced equation.

Exactly the same as Session 05 — the mole ratio doesn't care whether the reactants came from solids or solutions.

If both reactants are in solution, find the limiting reagent using the method from Session 06.

Step 3: Convert moles of target to the requested form.

Grams? $\text{mol} \times \text{molar mass}$.

Concentration? $\text{mol}/V_{\text{total}}(\text{L})$.

Volume of a solution needed? $V(\text{L}) = \text{mol}/M$.

$$\text{mL} \xrightarrow{\div 1000} \text{L} \xrightarrow{\times M} \text{mol A} \xrightarrow{\times \frac{\text{coeff B}}{\text{coeff A}}} \text{mol B} \xrightarrow{\times \text{MM}} \text{g B}$$

Copy this. The only new piece is the first arrow: $\text{mL} \rightarrow \text{L} \rightarrow \text{mol}$.

Pro Tips — Calculator Shortcuts & Mental Checks

Tip 1 — mL → L in your head. Move the decimal 3 places left. $25.0 \text{ mL} = 0.0250 \text{ L}$. $500.0 \text{ mL} = 0.5000 \text{ L}$. The #1 mistake in solution stoichiometry is forgetting this — it makes your answer 1000× too large.

Tip 2 — $M_1V_1 = M_2V_2$ works in mL. The units cancel, so you don't need to convert to liters: $2.00 \times 50.0 = M_2 \times 200.0 \rightarrow M_2 = 0.500 \text{ M}$. Only convert to L when you need actual moles ($\text{mol} = M \times V(\text{L})$).

Tip 3 — Dilution factor shortcut. Final volume $\div$ initial volume = dilution factor. $200/50 = 4 \rightarrow$ concentration drops by 4×. $2.00/4 = 0.500 \text{ M}$. Instant answer without the formula.

Tip 4 — Ion concentration = coefficient × compound concentration. In $CaCl_2$: $[Cl^-] = 2 \times [CaCl_2]$. In $Al_2(SO_4)_3$: $[Al^{3+}] = 2 \times [Al_2(SO_4)_3]$, $[SO_4^{2-}] = 3 \times [Al_2(SO_4)_3]$. This is always true for complete dissociation.

Tip 5 — Precipitate sanity check. If both solutions are dilute (~0.1 M) and volumes are tens of mL, your precipitate mass should be in the **milligram to single-digit gram** range. If you calculate 500 g of precipitate from two 50 mL solutions, you forgot the mL → L conversion.

Basic Drills (B1–B10)

Goal: Automate molarity, dilution, and simple solution stoichiometry.

- B1.** What is the molarity of a solution made by dissolving 11.7 g of NaCl in enough water to make 500.0 mL? (Na = 22.99, Cl = 35.45)
- B2.** How many moles of H_2SO_4 are in 25.0 mL of 0.500 M H_2SO_4 ?
- B3.** You dilute 100.0 mL of 6.00 M HCl to 500.0 mL. What is the new concentration?
- B4.** How many grams of NaOH are needed to prepare 250.0 mL of 0.200 M NaOH? (Na = 22.99, O = 16.00, H = 1.008)
- B5.** $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$. How many grams of AgCl form when 50.0 mL of 0.150 M $AgNO_3$ reacts with excess NaCl? (Ag = 107.87, Cl = 35.45)
- B6.** What is the $[Cl^-]$ in 0.300 M $CaCl_2$?
- B7.** (Dilution + stoichiometry) You dilute 25.0 mL of 2.00 M HCl to 100.0 mL. Then 20.0 mL of this diluted acid reacts with excess Zn. $Zn + 2HCl \rightarrow ZnCl_2 + H_2$. How many moles of H_2 are produced?
- B8.** What volume (in mL) of 0.500 M Na_2SO_4 contains 0.0250 mol of Na_2SO_4 ?
- B9.** (Two-part) Part 1: Find the molarity when 4.00 g of NaOH is dissolved in 200.0 mL. Part 2: This solution is then diluted to 500.0 mL. Find the new molarity. (NaOH = 40.00 g/mol)
- B10.** (Speed round — under 2 min) $BaCl_2 + Na_2SO_4 \rightarrow BaSO_4 + 2NaCl$. 30.0 mL of

0.200 M $BaCl_2$ (barium chloride) + excess Na_2SO_4 (sodium sulfate). Find grams of $BaSO_4$ (barium sulfate) precipitate. (Ba = 137.33, S = 32.07, O = 16.00)

Advanced Drills (A1–A10)

Goal: Precipitation stoichiometry with LR, dilution chains, and exam traps.

A1. (Multi-step with Session 06) 40.0 mL of 0.300 M $Pb(NO_3)_2$ + 60.0 mL of 0.250 M KI.
 $Pb(NO_3)_2 + 2KI \rightarrow PbI_2 + 2KNO_3$. (a) Find LR. (b) Grams of PbI_2 precipitate. (c) Grams of excess reagent remaining in solution. (Pb = 207.2, I = 126.90, K = 39.10, N = 14.01, O = 16.00)

A2. (Multi-step with Session 05) 25.0 mL of 0.400 M Na_2CO_3 + 35.0 mL of 0.300 M $CaCl_2$.
 $Na_2CO_3 + CaCl_2 \rightarrow CaCO_3 + 2NaCl$. (a) LR? (b) Grams $CaCO_3$? (c) What is the $[Na^+]$ in the final solution? (Assume volumes are additive.) (Na = 22.99, Ca = 40.08, C = 12.01, O = 16.00, Cl = 35.45)

A3. (Trap-laden) A student needs 0.0100 mol of HCl. They have 6.00 M HCl. They calculate: "V = 0.0100/6.00 = 0.00167 L = **0.167 mL**." What's wrong? Correct it.

A4. (Reverse engineering) A student dilutes a solution and calculates the new concentration as 0.045 M. Their partner gets 0.450 M — exactly 10× different. The original was 25.0 mL of 1.80 M diluted to 100.0 mL. Who is right? What mistake did the other make?

A5. (Constructive) Design a problem where two solutions are mixed, a precipitate forms, and the student must find both the mass of precipitate and the concentration of one ion remaining in solution. Solve it.

A6. (Constructive) Write a problem involving serial dilution (dilute, then dilute again). The final concentration should be exactly 0.0100 M. Solve it.

A7. (Table-based) Four solutions are prepared:

Solution	Solute	Mass (g)	Final Volume (mL)
A	NaCl	5.84	250.0

Solution	Solute	Mass (g)	Final Volume (mL)
B	$CaCl_2$	11.10	500.0
C	$C_6H_{12}O_6$	18.02	100.0
D	$Al_2(SO_4)_3$	34.22	200.0

(a) Calculate the molarity of each solution. (b) Calculate $[Cl^-]$ in solutions A and B. (c) Which solution has the highest total ion concentration? (Na = 22.99, Cl = 35.45, Ca = 40.08, C = 12.01, H = 1.008, O = 16.00, Al = 26.98, S = 32.07)

A8. (Timed — under 3 min) 20.0 mL of 0.500 M H_2SO_4 + 30.0 mL of 0.400 M NaOH.
 $H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O$. (a) LR? (b) Moles of Na_2SO_4 produced? (c) Which ion (H^+ or OH^-) is in excess, and what is its final concentration? Assume additive volumes. (H = 1.008, S = 32.07, O = 16.00, Na = 22.99)

A9. (Hardest variation) You have 500.0 mL of a solution containing both Na_2SO_4 (0.100 M) and NaCl (0.200 M). You add excess $Ba(NO_3)_2$. (a) What precipitate forms? (b) Calculate its mass. (c) After filtration, what ions and concentrations remain in solution? Assume no volume change. (Ba = 137.33, Na = 22.99, S = 32.07, O = 16.00, Cl = 35.45, N = 14.01)

A10. (Exam boss) A 1.000 g sample of impure Na_2CO_3 is dissolved in water and titrated with 0.500 M HCl. It takes 32.40 mL of HCl to reach the endpoint. $Na_2CO_3 + 2HCl \rightarrow 2NaCl + CO_2 + H_2O$. (a) How many moles of HCl were used? (b) How many moles of Na_2CO_3 were in the sample? (c) What mass of pure Na_2CO_3 does this correspond to? (d) What is the percent purity of the original sample? (Na = 22.99, C = 12.01, O = 16.00)

Now Read

You can now convert between concentration, volume, and moles; handle dilution; perform solution stoichiometry with limiting reagents; and calculate ion concentrations — at exam speed.

Read **Chapters 8 and 10** of *Everything You Need to Ace Chemistry in One Big Fat Notebook*.

You've solved 20 problems, from simple molarity to finding the percent purity of an impure sample.

Solutions are no longer mysterious — they're just moles dissolved in liters.

Solutions: <solutions/07-solutions.md>



Key Terms — Quick Reference

Term	What it means
molarity (M)	mol/L — the standard concentration unit in chemistry
solute	the substance being dissolved (e.g., salt)
solvent	the substance doing the dissolving (usually water)
dilution	adding solvent to lower concentration; $M_1 V_1 = M_2 V_2$
mL → L	divide by 1000 — the #1 mistake in solution problems
$M_1 V_1 = M_2 V_2$ in mL	volumes can stay in mL (they cancel); only convert when finding moles
precipitate	insoluble solid formed when two solutions are mixed
dissociation	ionic compound splits into ions: $NaCl \rightarrow Na^+ + Cl^-$
spectator ion	ion that doesn't participate in the reaction (e.g., Na^+ in many precipitations)
ion concentration	$[ion] = \text{coefficient} \times [compound]$ for complete dissociation